Worksheet 1

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Q. Automated Model Selection and Hyperparameter Tuning using Automated Machine Learning (AutoML).

I. Aim: To implement **Automated Machine Learning (AutoML)** using **TPOT** for **automated model selection and hyperparameter tuning**, and to evaluate the best machine learning pipeline for a given dataset.

II. Tasks to be done:

- Load and preprocess the dataset
- Split dataset into training and testing sets
- Apply AutoML using TPOT (Tree-Based Pipeline Optimization Tool)
- Automatically select the best ML algorithm
- Perform hyperparameter tuning automatically
- Limit search time using max_time_mins
- Train the best pipeline
- Evaluate model performance
- Analyze the effect of time constraint on AutoML

III. Dataset description:

Feature	Wine Dataset	Digits Dataset
Dataset Name	Wine Dataset	Digits Dataset
Input Features	Wine chemical properties - alcohol, acidity, magnesium, flavonoids, etc.	Handwritten digit images pixels (8×8 image = 64 features)
Number of Samples	(small-sized dataset)	(medium-sized dataset)
Number of Classes	Less classes (generally 3)	More classes(0–9, total 10)
Problem Type	Classification	Classification
Model Complexity	Simple data, models memorize easily.	Complex data, model learns general patterns
Overfitting Risk	High risk	Less risk
Accuracy Behavior	Accuracy 1.0 , high chances	Accuracy realistic (generally < 1.0)
Suitability for AutoML	Basic testing ke liye	Better evaluation and generalization.

IV. Code: 1> AutoMl code for Wine Dataset.

```
!pip install tpot

import joblib # Added import for joblib
from tpot import TPOTClassifier
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

```
# Load dataset
wine = load_wine()

x_train, x_test, y_train, y_test = train_test_split(
    wine.data, wine.target, test_size=0.2, random_state=42
)

# Change 1: Feature Scaling
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)

# Change 2: CV added
topt = TPOTClassifier(
    generations=3,
    population_size=25,
    cv=5,
    verbose=2, # Changed 'verbosity' to 'verbose'
    random_state=41,
    max_time_mins=2
)

topt.fit(x_train, y_train)

accuracy = topt.fitted_pipeline_.score(x_test, y_test)
print("Accuracy:", accuracy)

print("\nBest model selected by AutoML:")
print(topt.fitted_pipeline_)

# Use joblib to save the fitted pipeline
joblib.dump(topt.fitted_pipeline_, 'tpot_wine_quality_pipeline.pkl')
```

V. Output:

- **Without :** cv=5,verbose=2, random_state=41,max_time_mins=2
- **Accuracy : 1.**

- **With :** cv=5,verbose=2, random_state=41,max_time_mins=2

```
INFO:distributed.nanny:Nanny at 'tcp://127.0.0.1:43127' closed.
INFO:distributed.scheduler:Closing scheduler. Reason: unknown
INFO:distributed.scheduler:Scheduler closing all comms
Accuracy: 0.9444444444444444

Best model selected by AutoML:
Pipeline(steps=[('robustscaler',
                 RobustScaler(quantile_range=(0.1332202234918,
                                              0.8400801532853))),
                ('selectfwe', SelectFwe(alpha=0.0150626904778)),
                ('featureunion-1',
                 FeatureUnion(transformer_list=[('featureunion',
                                                 FeatureUnion(transformer_list=[('colu
                                                                                 Colum
                                                ('passthrough',
                                                 Passthrough())])),
                ('featureunion-2',
                 FeatureUnion(transformer_list=[('skiptransformer',
                                                 SkipTransformer()),
                                                ('passthrough',
                                                 Passthrough())])),
                ('randomforestclassifier'
```

VI. Code: 2> AutoMl code for Digit dataset.

```
!pip install tpot
import joblib # Added import for joblib
from tpot import TPOTClassifier
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
# Load dataset
wine = load_digits()
x_train, x_test, y_train, y_test = train_test_split(
```

```
wine.data, wine.target, test_size=0.2, random_state=42
)# Change 1: Feature Scaling
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
# Change 2: CV added
topt = TPOTClassifier(
    generations=3,
    population_size=25,
    cv=5,
    verbose=2, # Changed 'verbosity' to 'verbose'
    random_state=41,
    max_time_mins=2
)
topt.fit(x_train, y_train)
accuracy = topt.fitted_pipeline_.score(x_test, y_test)
print("Accuracy:", accuracy)
print("\nBest model selected by AutoML:")
print(topt.fitted_pipeline_)
# Use joblib to save the fitted pipeline
joblib.dump(topt.fitted_pipeline_, 'Digit')
```

Output: The accuracy difference with and without using parameters like cv=5, random_state, and max_time_mins=2 is very small**.**

```
INFO:distributed.scheduler:Retire worker addresses (stimulus_id='retire-workers-1767887223.7386453') (0,)
INFO:distributed.nanny:Closing Nanny at 'tcp://127.0.0.1:39449'. Reason: nanny-close
INFO:distributed.nanny:Nanny asking worker to close. Reason: nanny-close
INFO:distributed.core:Received 'close-stream' from tcp://127.0.0.1:48332; closing.
INFO:distributed.scheduler:Remove worker addr: tcp://127.0.0.1:36767 name: 0 (stimulus_id='handle-worker-cleanup-176
ERROR:distributed.scheduler:Removing worker 'tcp://127.0.0.1:36767' caused the cluster to lose scattered data, which
INFO:distributed.scheduler:Lost all workers
INFO:distributed.nanny:Nanny at 'tcp://127.0.0.1:39449' closed.
INFO:distributed.scheduler:Closing scheduler. Reason: unknown
INFO:distributed.scheduler:Scheduler closing all comms
Accuracy: 0.9527777777777777

Best model selected by AutoML:
Pipeline(steps=[('standardscaler', StandardScaler()),
                ('selectfwe', SelectFwe(alpha=0.0123744574969)),
                ('featureunion-1',
                 FeatureUnion(transformer_list=[('featureunion',
                                                 FeatureUnion(transformer_list=[('nystroem',
                                                                                 Nystroem(gamma=0.6703535391704,
                                                                                          kernel=np.str_('cosine'),
                                                                                          n_components=29,
                                                                                          random_state=41))])),
```

VII. Code Summary & Important Points :

Aspect	Description
Programming Language	Python
Libraries Used	TPOT, scikit-learn, joblib
Dataset Used	Wine Dataset , digit dataset
Problem Type	Multi-class Classification
Input Features	Wine chemical properties (numerical features)
Target Variable	Wine class labels
Dataset Preprocessing	Feature scaling performed using StandardScaler to normalize input features
Data Splitting	Dataset split into 80% training and 20% testing using train_test_split
Model Approach	AutoML using TPOTClassifier
Generations	3 (number of evolutionary iterations)
Population Size	25 candidate pipelines per generation
Cross-Validation	5-fold cross-validation (cv = 5) used during model selection
Time Constraint	AutoML search limited using max_time_mins = 2
Model Training	TPOT automatically trains and evaluates multiple ML pipelines

Aspect	Description
Best Model Selection	Best performing pipeline is selected automatically by TPOT
Evaluation Metric	Accuracy score on the test dataset
Model Output	Trained classification pipeline
Model Saving	Final pipeline saved using joblib for future use

VIII. Process table :

Step No.	Process Description
Step 1	Initially, AutoML was applied on the Wine dataset without proper constraints, which resulted in accuracy = 1.0, indicating possible overfitting due to small dataset size.
Step 2	To improve model reliability, feature scaling using StandardScaler was applied to normalize input features.
Step 3	5-fold cross-validation (cv = 5) was introduced to evaluate models on multiple data splits and improve generalization.
Step 4	max_time_mins = 2 was used to limit AutoML search time and prevent overly complex pipelines.
Step 5	The same tuned parameters were then applied to a larger and more complex Digits dataset to better evaluate AutoML performance.
Step 6	Using the Digits dataset resulted in realistic accuracy and reduced overfitting, showing improved model generalization and stability.

IX. Learning Outcomes (What I have learnt):

- Understood the concept of **Automated Machine Learning (AutoML)**
- Learned how TPOT automates model selection
- Observed automatic hyperparameter tuning
- Understood the role of max_time_mins in controlling search complexity
- Learned how AutoML reduces manual effort in ML workflows
- Gained practical experience in evaluating AutoML-generated models